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| TECHNICAL PRINCIPLES | <b>7.2 How can materials and processes be used to make final prototypes?</b>                                                                                                                                                                                                                                                                                                                                                                      |                                                                                     |
|                      | <p>a. Understand how to select and safely use common workshop tools, equipment and machinery to manipulate materials by methods of:</p> <ul style="list-style-type: none"> <li>i. wasting/subtraction processes such as cutting, drilling, turning, milling</li> <li>ii. addition processes such as soldering, brazing, welding, adhesives, fasteners</li> <li>iii. deforming and reforming processes such as bending, vacuum forming.</li> </ul> |  |
|                      | <p>b. Demonstrate an understanding of the role of computer-aided manufacture (CAM) and computer-aided engineering (CAE) to fabricate parts of a final prototype:</p> <ul style="list-style-type: none"> <li>i. additive manufacturing (3D printing) to fabricate a usable part</li> <li>ii. subtractive CNC manufacturing such as laser/plasma cutting, milling, turning and routing.</li> </ul>                                                  |                                                                                     |
|                      | <p>c. Demonstrate an understanding of measuring instruments and techniques used to ensure that products are manufactured accurately or within tolerances as appropriate.</p>                                                                                                                                                                                                                                                                      |  |
|                      | <p>d. Understand how the available forms, costs and working properties of materials contribute to the decisions about suitability of materials when developing and manufacturing their own products.</p>                                                                                                                                                                                                                                          |  |